

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE					
Regular and Supplementary Winter 2024					
Course: B.Tech.		Branch : Information Technology		Semester :VII	
Subject Code & Name: BTITPE705B & Information Security					
Max Marks: 60		Date: 21/02/2025		Duration: 3 Hr.	
Instructions to the Students:					
1. Each question carries 12 marks.					
2. Question No. 1 will be compulsory and include objective-type questions.					
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.					
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.					
5. Use of non-programmable scientific calculators is allowed.					
6. Assume suitable data wherever necessary and mention it clearly.					
				(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)				12
1	What are the primary components of computer security concepts?				1
	a. Threats, Attacks, and Solutions	b. Threats, Attacks, and Assets	c. Risks, Management, and Policies	d. Data, Systems, and Security	
	Which cryptographic tool uses a pair of keys – public and private?				1
	a. Symmetric encryption	b. Hashing	c. Public Key Encryption	d. Malware protection	
3	What are the primary types of attacks in computer security?				1
	a. Symmetric and Asymmetric	b. Active and Passive	c. Data-based and Encryption-based	d. Preventive and Corrective	
4	Which malware type specifically captures keystrokes?				1
	a. Phishing software	b. Keylogger	c. Trojan	d. Worm	
5	Which of the following ensures data integrity and authenticity?				1
	a. Spyware countermeasures	b. Symmetric encryption	c. Digital signatures	d. Hash functions	
	Federated databases are designed to:				1
	a. Integrate malware countermeasures	b. Manage distributed databases securely	c. Enhance single database security	d. Improve network access speed	
7	Which of the following is considered a security threat?				1
	a. Authentication system	b. Malicious software	c. Network firewall	d. Public key encryption	
8	Which of the following provides message authentication?				1
	a. Stealth malware	b. Hash functions	c. Public Key Infrastructure (PKI)	d. Symmetric encryption	

9	What is the main purpose of a digital signature?					1
	a. To encrypt the message content	b. To authenticate the sender and ensure message integrity	c. To mask sensitive data	d. To manage symmetric encryption keys		
10	What does SSE-CMM stand for?					1
	a. Secure System Evaluation-Cyber Management Model	b. System Security Engineering-Capability Maturity Model	c. Security Standards Evaluation-Control Management Model	d. System Security Evaluation and Control		
11	Which access control model allows discretionary access permissions?					1
	a. Mandatory Access Control	b. Discretionary Access Control	c. Role-Based Access Control	d. Attribute-Based Access Control		
12	What privacy issue arises in smart card applications?					1
	a. Unauthorized access to sensitive user data	b. Limited storage capabilities	c. Delayed transaction processing	d. Reduced encryption strength		
Q.3	Solve the following.					12
	Describe symmetric encryption with a suitable diagram.					6
	Explain Caesar Cipher with example.					6
Q.3	Solve the following.					12
A)	Define access control. What are its primary goals? Explain with diagram.					6
B)	What is attack? Explain its types with neat diagram.					6
Q. 4	Solve Any Two of the following.					12
A)	Explain the concept of Message Authentication with diagram.					6
B)	Define role-based access control (RBAC). How is it different from discretionary access control (DAC)?					6
	What are the key stages in the policy design life cycle?					6
Q.3	Solve Any Two of the following.					12
A)	Discuss various countermeasures that can be employed to protect systems from malware attacks.					6
B)	What are digital signatures? Explain DSS algorithm with neat diagram?					6
C)	Explain the PDCA model with its application in security policy design.					6

Q. 6	Solve Any Two of the following.		12
A)	What is ethical hacking? How can it be used to improve security?		6
B)	What is network security? Differentiate between trusted and untrusted networks.		6
C)	What is data masking? How can it be used to protect privacy in outsourcing?		6
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